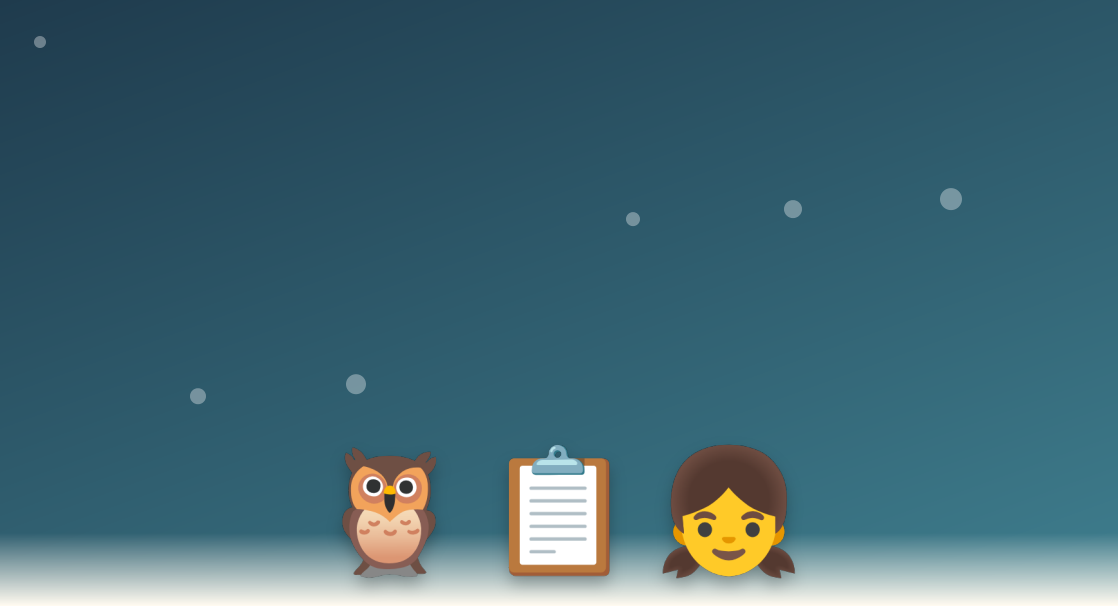


The Great Kingdom Census

I PUC · Ch 2 · Biological Classification

Starring Bindu 👧 & Prof. BAIO 🦉

A BAIO storybook by Narendra Sir · Biology Lecturer



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The Census Begins

Prof. BAIO pinned a giant chart on the lab wall. "Bindu, every living thing needs a home address. In 1969, **R.H. Whittaker** sorted all life into five kingdoms." Bindu read them aloud: **Monera**, **Protista**, **Fungi**, **Plantae** and **Animalia**. "Today," said the owl, "we visit them all!"

💡 REMEMBER

Whittaker's five kingdoms are based on cell structure, body organisation, mode of nutrition, reproduction and phylogenetic relationships.

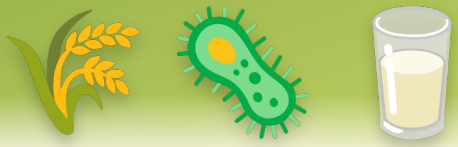


Kingdom Monera

The first stop was the kingdom of the tiniest. "Everyone here is a **prokaryote** — no nuclear membrane!" hooted Prof. BAIO. In a boiling hot spring lived the **thermoacidophiles**. In salty pans lived the **halophiles**. And inside a cow's stomach, the **methanogens** were busy making methane gas.

💡 **REMEMBER**

Archaebacteria survive extreme places because of a special cell wall. Methanogens produce methane — biogas.

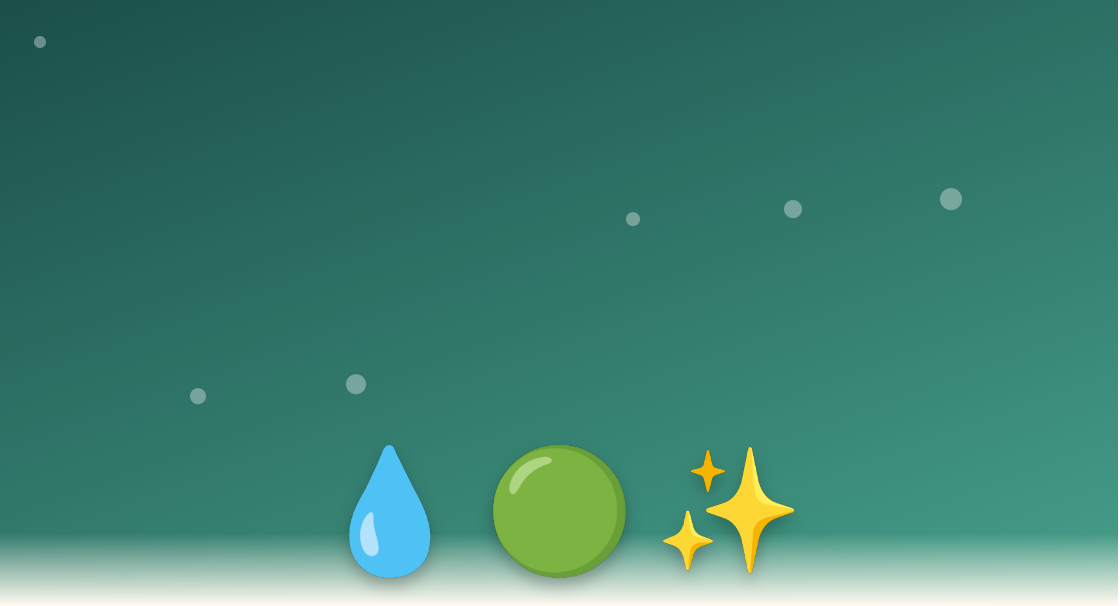


The Useful Bacteria

Next came the **eubacteria**, the true bacteria. Green **cyanobacteria** like Nostoc and Anabaena waved from a paddy field — they fix nitrogen in special cells called **heterocysts**. Then the smallest of all, **Mycoplasma**, peeped out shyly. "I have no cell wall at all!"

💡 **REMEMBER**

Mycoplasma has no cell wall and is the smallest living cell; it can survive without oxygen. Bacteria reproduce mainly by fission.



The Protist Pond

In a sparkling pond lived the single-celled eukaryotes of **Protista**. **Diatoms** glittered in glassy silica shells; their piled-up walls form **diatomaceous earth**. **Euglena** made food in sunlight but hunted in the dark. Far out at sea, **dinoflagellates** like Gonyaulax multiplied so fast that the water turned red — a red tide!

💡 **REMEMBER**

Diatoms are the chief producers in the oceans. Euglena has a protein-rich pellicle instead of a cell wall.



The Protozoan Parade

Four kinds of **protozoans** marched past. Amoeba crawled on **pseudopodia**. Trypanosoma whipped its flagellum — it causes sleeping sickness. Paramecium rowed with tiny **cilia**. Last came **Plasmodium**, a sporozoan, who whispered, "I cause malaria."

💡 REMEMBER

Protozoans: amoeboid (Amoeba), flagellated (Trypanosoma), ciliated (Paramecium) and sporozoans (Plasmodium).



The Mushroom Mansion

The Fungi kingdom smelled of bread and rain. Their walls were made of **chitin**, and their bodies were long threads called **hyphae** woven into a **mycelium**. Rhizopus grew on old bread, Puccinia gave wheat its rust, and Penicillium gave the world an antibiotic.

💡 **REMEMBER**

Fungi are heterotrophs. Classes: Phycomycetes, Ascomycetes, Basidiomycetes and Deuteromycetes (the "imperfect fungi").



Not Quite Alive

In a dark corner hid three puzzles. A **virus** was just a nucleoprotein, lifeless outside a host — Ivanowsky discovered one in tobacco in 1892. A **viroid** was naked RNA with no protein coat. A **prion** was a wrongly folded protein that causes mad cow disease.

💡 **REMEMBER**

Viruses are not placed in any of the five kingdoms. Viroids were discovered by T.O. Diener (1971) — potato spindle tuber disease.



The Best Friends

On a rock, an alga and a fungus waved together. "We are a **lichen**," said the alga. "I make the food — I am the **phycobiont**." "And I give shelter and water — I am the **mycobiont**," said the fungus. Prof. BAIO smiled: "Remember, lichens never grow where the air is polluted."

💡 **REMEMBER**

Lichens are a mutual partnership of algae and fungi — they are good indicators of air pollution.



KEY TERMS

Words you met in this story

Tap a word to jump to its page. Can you explain each one?

R.H. Whittaker p.1 · Monera p.1 · Protista p.1 · Fungi p.1 · Plantae p.1 · Animalia p.1 · prokaryote p.2 · thermoacidophiles p.2 · halophiles p.2 · methanogens p.2 · eubacteria p.3 · cyanobacteria p.3 · heterocysts p.3 · Mycoplasma p.3 · Diatoms p.4 · diatomaceous earth p.4 · Euglena p.4 · dinoflagellates p.4 · protozoans p.5 · pseudopodia p.5 · cilia p.5 · Plasmodium p.5 · chitin p.6 · hyphae p.6 · mycelium p.6 · virus p.7 · viroid p.7 · prion p.7 · lichen p.8 · phycobiont p.8 · mycobiont p.8



QUICK QUIZ

Do you remember?

1. Who proposed the five kingdom classification?

- Linnaeus
- R.H. Whittaker
- Darwin
- Haeckel

2. Which organism has no cell wall?

- Nostoc
- Mycoplasma
- Euglena
- Diatom

3. Methanogens produce:

- Oxygen
- Methane
- Nitrogen
- Ammonia

4. The cell wall of fungi is made of:

- Cellulose
- Chitin
- Silica
- Peptidoglycan

5. A lichen is an association of:

- Alga and fungus
- Bacterium and fungus
- Alga and bacterium
- Fungus and root

Answers: 1-R.H. Whittaker · 2-Mycoplasma · 3-Methane · 4-Chitin · 5-Alga and fungus